US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250274114
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	Santillan; Jofrey G. et al.

SOLID STATE SWITCH AND A CIRCUIT

Abstract

A solid state switch, comprising a first field-effect transistor (FET). The first FET has a first terminal, a second terminal, a bulk terminal and a gate terminal, and is configured to be switched between an on-state and an off-state. The solid state switch also comprises a second FET in series with the first FET. The second FET has a first terminal, a second terminal, a bulk terminal, and a gate terminal. The second terminal of the first FET is connected to the second terminal of the second FET. The solid state switch comprises a first buffer comprises an output terminal coupled to the bulk terminal of the first FET, and an input terminal coupled to the first terminal of the first FET.

Inventors: **Santillan; Jofrey G. (Gen. Trias, PH), McDonagh; Declan (Limerick, IE), Aherne; David (Limerick City, IE)**

Applicant: **Analog Devices International Unlimited Company** (Limerick, IE)

Family ID: **1000008376618**

Appl. No.: **19/007325**

Filed: **December 31, 2024**

Related U.S. Application Data

parent US continuation-in-part 18586378 20240223 PENDING child US 19007325

Publication Classification

Int. Cl.: **H03K17/08** (20060101)

U.S. Cl.:

CPC **H03K17/08** (20130101);

Background/Summary

CLAIM OF PRIORITY [0001] This application is a Continuation-In-Part of U.S. patent application Ser. No. 18/586,378, filed Feb. 23, 2024, the benefit of priority which is claimed hereby, and which is incorporated by reference herein in its entirety.

FIELD
[0002] This application relates to compensating for leakage in a semiconductor switch (i.e. a solid state switch). Specifically, a field-effect-transistor (FET), e.g., a JFET, Silicon Carbide FET, or a low voltage FET, such as a FET with an accessible bulk (also called ‘body’, or ‘back-gate’) terminal.
BACKGROUND
[0003] Solid state switches may be used with components such as, a precision measurement apparatus, and automated test equipment. A solid state switch can have an associated leakage which affects the results at a precision measurement apparatus. For example, a large leakage current can reduce power efficiency and reduces the accuracy of component measurements connected via a solid state switch. Leakage reduction herein refers to reducing the current leakage at an input and/or output terminal of a solid state switch.
[0004] A Junction FET (JFET) is a type of transistor that controls current flow through an electric field applied to a gate. JFETs comprise a channel of either N-type or P-type semiconductor material, with the current flow controlled by a voltage applied to the gate, which forms a p-n junction with the channel. JFETs are characterized by their high input impedance and low noise, making them suitable for applications in analog signal processing and amplification.
[0005] A Silicon Carbide FET (SIC) is a type of transistor that employs silicon carbide as the semiconductor material. SiC's high breakdown voltage allows efficient operation at high temperatures and voltages compared to traditional silicon-based transistors.
SUMMARY OF THE DISCLOSURE
[0006] The present disclosure provides a solid state switch device which compensates for current leakage. In particular, the examples herein provide leakage compensation techniques which are suitable with non-isolated and fully isolated (e.g., a silicon-on-insulator (SOI), or buried oxide isolated device) switches. In addition, the examples herein also provide an area efficient implementation. Such solid state switch devices are suitable for use with AC and/or DC voltage precision measurement apparatuses and AC and/or DC automated test equipment.
[0007] According to a first aspect there is provided a solid state switch, comprising: [0008] a first field-effect transistor, FET, comprising: a first terminal, a second terminal, a bulk terminal and a gate terminal, and configured to be switched between an on-state and an off-state; [0009] a second FET in series with the first FET, wherein the second FET comprises a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first FET is connected to the second terminal of the second FET; and, [0010] a first buffer comprising an output terminal coupled to the bulk terminal of the first FET, and an input terminal coupled to the first terminal of the first FET.
[0011] Optionally, the first terminal is a drain terminal. Optionally, the second terminal is a source terminal. Optionally, the first buffer is arranged to provide a current source to the bulk terminal of the first FET.
[0012] Optionally, the first buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.
[0013] Optionally, the first aspect further comprises a second buffer comprising an output terminal coupled to the bulk terminal of the second FET, and an input terminal coupled to the first terminal of the second FET. Optionally, the second buffer is arranged to provide a current source to the bulk

terminal of the second FET.
[0014] Optionally, the second buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.
[0015] Optionally, the first buffer is a UGB so as to reduce leakage at the first terminal of the first FET.
[0016] Optionally, the first FET is a low voltage FET, and optionally the second FET is a low voltage FET. Optionally, a low voltage FET is a FET arranged to be used with voltage differences of less than or equal to 5V, i.e., voltage rated at 5V.
[0017] Optionally, the first FET is an isolated FET, and the second FET is an isolated FET.
[0018] Optionally, the first terminal of the first FET is a drain terminal. Optionally, the first terminal of the second FET is a drain terminal. Optionally, the second terminal of the first FET is a source terminal. Optionally, the second terminal of the second FET is a source terminal.
[0019] Optionally, the first and second FET are both a first-type FET. Optionally, the first-type is n-type or p-type.
[0020] Optionally, the first aspect further comprises an electrical component for overvoltage protection. Optionally, the electrical component is arranged in series between the first and second FETs.
[0021] Optionally, the electrical component is a third FET. Optionally, the third FET is a second-type FET. Optionally, the second-type is p-type or n-type. Optionally, the first-type is different to the second-type. Optionally, the first, second, and third FETs have the same voltage rating.
[0022] Optionally, the electrical component protects the solid state switch in applications where overvoltage conditions can cause damage to a device to which the switch is connected.
[0023] Optionally, the solid state switch of the first aspect is a transmission-gate switch comprising the first FET in parallel with a third FET. Optionally, the third FET, comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the third FET is configured to be switched between an on-state and an off-state. Optionally, the third FET is a first-type FET. Optionally, the first FET is a second-type FET. Optionally, the first-type is n-type or p-type. Optionally, the second-type is p-type or n-type. Optionally, the first-type is different to the second-type.
[0024] Optionally, the first terminal of the third FET is coupled to the first terminal of the first FET, and the second terminal of the third FET is coupled to the first terminal of the second FET.
[0025] Optionally, the first terminal of the third FET is coupled to the first terminal of the first FET, and the second terminal of the third FET is coupled to the second terminal of the first FET.
[0026] Optionally, the solid state switch of the first aspect further comprises a fourth FET in parallel with the second FET. Optionally, the fourth FET comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the fourth FET is configured to be switched between an on-state and an off-state. Optionally, the fourth FET is a third-type FET. Optionally, the second FET is a fourth-type FET. Optionally, the third-type is n-type or p-type. Optionally, the fourth-type is p-type or n-type. Optionally, the third-type is different to the fourth-type.
[0027] Optionally, the fourth FET is in series with the third FET. Optionally, the second terminal of the third FET is connected to the second terminal of the fourth FET.
[0028] Optionally, the output terminal of the first buffer is coupled to the bulk terminal of the third FET, and the input terminal of the first buffer is coupled to the first terminal of the third FET.
[0029] Optionally, the output terminal of the second buffer is coupled to the bulk terminal of the fourth FET, and the input terminal of the second buffer is coupled to the first terminal of the fourth FET.
[0030] Optionally, the first terminal of the third FET is a drain terminal. Optionally, the first terminal of the fourth FET is a drain terminal. Optionally, the second terminal of the third FET is a source terminal. Optionally, the second terminal of the fourth FET is a source terminal.
[0031] Optionally, the third FET is a low voltage FET, and optionally the fourth FET is a low

voltage FET.
[0032] Optionally, the third FET is an isolated FET, and optionally the fourth FET is an isolated FET.
[0033] Optionally, the first FET is a JFET or a Silicon Carbide FET (SIC). Optionally, the second FET is a JFET or a SiC. Optionally, the third FET is a JFET or a SiC. Optionally, the fourth FET is a JFET or a SiC.
[0034] According to a second aspect there is provided a solid state switch, comprising: [0035] a first field-effect transistor, FET, comprising: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and configured to be switched between an on-state and an off-state; [0036] a second FET in series with the first FET, wherein the second comprises: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first FET is coupled to the second terminal of the second FET; [0037] means for providing a current source to the bulk terminal of the first FET based on the voltage at the first terminal of the first FET; and [0038] means for providing a current source to the bulk terminal of the second FET based on the voltage at the first terminal of the second FET.
[0039] Optionally, the means for providing a current source to the bulk terminal of the first FET based on the voltage at the first terminal of the first FET is one or more first electronic components.
[0040] Optionally, means for providing a current source to the bulk terminal of the second FET based on the voltage at the first terminal of the second FET is one or more second electronic components.
[0041] Optional features of the first aspect may be applied to the second aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0042] FIG. **1**_a_ illustrates a design of an n-type MOSFET switch with its parasitic diodes D**1**_a_ and D**1**_b_.
[0043] FIG. **1**_b_ illustrates a source of dominant current leakage through the MOSFET switch of FIG. **1**_a_.
[0044] FIG. **2** illustrates an example of a semiconductor structure of an isolated MOSFET switch, with its parasitic diodes D**1**_a_ and D**1**_b_.
[0045] FIG. **3** illustrates a bi-directional solid state switch with unidirectional leakage reduction comprising a first n-type MOSFET coupled in series to a second n-type MOSFET, the first MOSFET being coupled to a buffer to generate a bulk voltage substantially equal to its drain voltage.
[0046] FIG. **4**_a_ illustrates a bidirectional solid state switch comprising a first n-type MOSFET coupled in series to a second n-type MOSFET, each MOSFET being coupled to a buffer to generate a bulk voltage substantially equal to its drain voltage.
[0047] FIG. **4**_b_ illustrates a bidirectional solid state switch comprising a first p-type MOSFET coupled in series to a second p-type MOSFET, each MOSFET being coupled to a buffer to generate a bulk voltage substantially equal to its drain voltage.
[0048] FIG. **5**_a_ illustrates a bi-directional solid state T-gate switch with bidirectional leakage reduction.
[0049] FIG. **5**_b_ illustrates a bi-directional solid state T-gate switch with unidirectional leakage reduction.
[0050] FIG. **5**_c_ illustrates a bi-directional solid state T-gate switch with partial leakage reduction.
[0051] FIG. **6**_a_ illustrates a bi-directional solid state switch with bidirectional leakage reduction with overvoltage protection circuitry.
[0052] FIG. **6**_b_ illustrates a bi-directional solid state switch with bidirectional leakage reduction

with overvoltage protection circuitry.
[0053] FIG. **7***a* illustrates a bi-directional solid state switch with unidirectional leakage reduction comprising a first n-type FET coupled in series to a second n-type FET, the first FET being coupled to a buffer to generate a bulk voltage substantially equal to its drain voltage.
[0054] FIG. **7***b* illustrates a bi-directional solid state switch with unidirectional leakage reduction comprising a first JFET coupled in series to a second JFET, the first JFET being coupled to a buffer to generate a bulk voltage substantially equal to its drain voltage.
[0055] FIG. **8** illustrates a bi-directional solid state T-gate switch with bidirectional leakage reduction.
[0056] FIG. **9** illustrates a bi-directional solid state switch with bidirectional leakage reduction with overvoltage protection circuitry.

DETAILED DESCRIPTION

[0057] A metal-oxide-semiconductor field-effect-transistor (MOSFET or MOS) device when configured to function as a switch is susceptible to parasitic capacitance and current leakage which can negatively impact measurement performance of equipment connected via said MOS device. To overcome the impact of parasitic capacitance and current leakage, additional circuitry can be coupled to the MOS device configured to function as a switch. Most types of field-effect-transistor devices with a designer accessible bulk terminal can benefit from the additional circuitry. For example, the devices described below are MOSFET devices configured to function as switches. A MOS device may imply a low-voltage device. For example, low voltage may be defined as: less than or equal to 5V, or, less than or equal to 10V. A low voltage MOS may be configured to operate with a Vds which is less than or equal to 5V, or, less than or equal to 10V.
[0058] As a brief non-limiting overview of the invention, a new MOS based switch for use in/with precision instruments is provided. The switch can enable leakage compensation. That is reduced current leakage at the input and/or output of the switch. The new MOS based switch can comprise a MOS device coupled to another MOS device in series and a respective buffer coupled to a bulk terminal of each MOS device for providing a current source to the bulk terminal of each MOS device, such that the new MOS based switch appears not to leak current from the perspective of the input and/or output terminal of the new MOS based switch. The buffer may comprise a unity gain buffer (UGB); a voltage follower; or a cascade complementary source follower.
[0059] FIG. **1***a* shows an n-type MOS **10** (i.e., NMOS **10**) with its parasitic diodes D**1***a* and D**1***b*. The parasitic diodes D**1***a* and D**1***b* of the NMOS **10** result from the fabrication process and are present in the majority of types of MOS switches. The parasitic diodes D**1***a* and D**1***b* are formed between P-type and N-type material of the NMOS **10**. The parasitic diode D**1***a* of the NMOS **10** is formed between the p-well and a source n-type layer. The parasitic diode D**1***b* of the NMOS **10** is formed between the drain n-type and the p-well layer. If the NMOS device was a non-isolated MOS device, then the p-well layer may be the P-type substrate layer.
[0060] NMOS **10** comprises a gate terminal **11**, a drain terminal **12**, a source terminal **14**, and a bulk terminal **16**. The bulk terminal **16** is accessible to a circuit designer. In prior-art applications, the bulk terminal is directly electrically coupled (and therefore, biased) to the source terminal to avoid body effect.
[0061] A P-type MOS (PMOS) (or other p-type FET switch) could be described similarly.
[0062] FIG. **1***b* shows the NMOS **10** of FIG. **1***a* with a leakage current shown. When the NMOS **10** is in the on-state (i.e., switched 'ON') the drain/source voltage can be any value usually between the power supplies (e.g., Vdd and Vss). If the NMOS **10** is switched 'ON', the voltage difference between the source terminal **14** and the drain terminal **12** is approximately 0V (assuming a negligible/low ON-resistance). Typically, the bulk terminal **16** is coupled to the source terminal **14**, thus, parasitic diode D**1***a* may be ignored.
[0063] If the NMOS **10** is switched 'OFF' and the voltage at the drain **12** (e.g., 5V, or the most positive supply, e.g., Vdd) is greater than the voltage at the source **14** (e.g., 0V/gnd, or the most

negative supply, e.g., Vss), then there is a large voltage differential across the parasitic diode D**1***b*. The large voltage differential across the parasitic diode D**1***b* causes a leakage current i.sub.lkg_off through the reverse biased parasitic diode D**1***b*. The leakage current i.sub.lkg_off reduces power efficiency and reduces the accuracy of component measurements connected via the NMOS **10**. The leakage current may be defined by the Shockley diode equation:

[00001] $i_{\mathrm{lkg\_off}} = i_s(e^{\frac{V_d}{nV_t}} - 1)$ (1) [0064] where i.sub.s is the reverse-bias saturation current (or scale current), V.sub.d is the voltage across the parasitic diode D**1***b*, V.sub.t is the thermal voltage, and n is the ideality factor (also known as the quality factor or emission coefficient) and depends on the fabrication process and semiconductor material. Therefore, if the voltage across the parasitic diode D**1***b* is non-zero, then the leakage current i.sub.lkg_off is large.

[0065] FIG. **2** shows an example of a semiconductor structure of an NMOS transistor, specifically an isolated NMOS **10**. The parasitic diodes D**1***a* and D**1***b* are shown in FIG. **2**. Note that the P-well is accessible to the circuit designer via the bulk terminal **16**.

[0066] As shown in FIGS. **1***a*, **1***b*, and **2**, the NMOS **10** is susceptible to leakage currents which reduce the accuracy of component measurements connected via the NMOS **10**. Leakage current can also reduce the power efficiency of the NMOS **10** or components connected via the NMOS **10**.

[0067] FIG. **3** shows a bi-directional solid state switch **20***a* with unidirectional leakage reduction solid state switch **20***a* comprising the first NMOS **10** coupled to a first buffer **36**. The first NMOS **10** being configured to be switched between an on-state (i.e., switched 'ON') and an off-state (i.e., switched 'OFF'). The first buffer **36** includes an output terminal **37**; and an input terminal **38** coupled to the drain terminal **12** of the first NMOS **10**. The first buffer **36** is arranged to provide a voltage to the bulk terminal **16** substantially equal to the voltage at the drain terminal **12** of the first NMOS **10**. This reduces the leakage current i.sub.lkg_off through parasitic diode D**1***b* as per equation (1). The first buffer **36** connected to the first NMOS **10** may also be arranged to provide a current source to the bulk terminal **16** of the first NMOS **10** based on the voltage at the drain terminal **12** of the first NMOS **10**. Thus, the addition of the first buffer **36** reduces leakage via the parasitic diode D**1***b* when compared to an arrangement of the first NMOS **10** where the bulk terminal **16** is directly electrically coupled to the source terminal **14** (e.g., without the first buffer **36** present), or a negative voltage supply terminal (e.g., gnd, Vss). Therefore, the first buffer **36** may reduce (or eliminate) current leakage at the drain terminal **12** of the first NMOS **10** when the voltage at the drain terminal **12** of the first NMOS **10** is greater than the voltage at the source terminal **14** of the first NMOS **10** (e.g., by providing a 0V across the parasitic diodes D**1***b*). However, the first buffer **36** may act as a current source and provide a large leakage current over parasitic diode D**1***a* to the source terminal **14** of the first NMOS **10**, if the voltage at the bulk terminal **16** of the first NMOS **10** is greater than the voltage at the source terminal **14** of the first NMOS **10**.

[0068] To prevent the large leakage current over parasitic diode D**1***a* to the source terminal **14** of the first NMOS **10** leaking through the solid state switch when it is in an off-state **10**, a mirrored arrangement of the first NMOS **10** is provided by a second NMOS **40**. The first NMOS **10** may be substantially similar to the second NMOS **40**. Thus, when the voltage at the drain terminal **12** of the first NMOS **10** is greater than the voltage at the drain terminal **42** of the second NMOS **40**, current leakage through the solid state switch **20***a* when it is in an off-state may be reduced (or eliminated). The solid state switch **20***a* is constructed by coupling the source terminals **14**, **44** of each first NMOS **10** and second NMOS **40** together (either directly or optionally, via an electrical component for overvoltage protection). This arrangement ensures that the parasitic diode D**2***a* (between a source terminal **44** and a bulk terminal **45** of the second NMOS **40**) may be reverse biased and reduces the leakage current sourced from the first buffer **36** over the forward biased parasitic diode D**1***a*. There may be a small current leakage via the reverse biased parasitic diode D**2***a* (i.e., sourced from the first buffer **36**), however, advantageously, this is not 'seen' at (i.e., does not interact with) the drain terminal of the first NMOS **10**. Thus, from the perspective of the input terminal **12** of the solid state switch **20***a*, current leakage may be eliminated.

[0069] Specifically, FIG. **3** shows the solid state switch **20***a* comprises the second NMOS **40**. The second NMOS **40** being configured to be switched between an on-state and an off-state. FIG. **3** shows the second NMOS **40** with parasitic diodes D**2***a* and D**2***b*. The second NMOS **40** comprises a gate terminal **41**, a drain terminal **42**, a source terminal **44**, and a bulk terminal **45**. The bulk terminal **45** is accessible to a circuit designer.

[0070] An input of the solid state switch **20***a* may be the drain terminal **12** of the first NMOS **10**. An output of the solid state switch **20***a* may be the drain terminal **42** of the second NMOS **40**.

[0071] A switching signal coupled to the gate **41** of the second NMOS **40** may also be coupled to the gate **11** of the first NMOS **10**, such that the first and second NMOSs **10**, **40** are configured to be in the same switching state, either 'ON' or 'OFF'. Thus, during an off-state of the solid state switch **20**, the first and second MOSs **10** and **40** can be switched 'OFF' simultaneously. Similarly, during an on-state of the solid state switch **20**, the first and second MOSs **10** and **40** can be switched 'ON' simultaneously.

[0072] In an off-state, the bulk terminal **45** if second NMOS **40** may be coupled to the most negative supply of the switch (e.g., 0V, gnd, or Vss). In an on-state, the bulk terminal **45** of second NMOS **40** may also be coupled to the most negative supply of the switch (e.g., 0V, gnd, or Vss). Alternatively, in an on-state, the bulk terminal **45** of second NMOS **40** may be back gate switched. In back gate switching, the bulk terminal (e.g., **45**) is tied to the signal passing through the switch (e.g., second NMOS **40**) to reduce the body effect and improve performance.

[0073] The solid state switch **20***a* of FIG. **3** may be particularly advantageous when used in a multiplexed system.

[0074] FIG. **4***a* shows a bi-directional solid state switch **20***b* with bi-directional leakage reduction comprising the solid state switch **20***a* of FIG. **3**, and a second buffer **46**. That is, the arrangement of the second NMOS **40** and the second buffer **46***a* is a mirrored arrangement of the first NMOS **10** and the first buffer **36**. The first buffer **36** may be substantially similar to the second buffer **46**. The second buffer **46** may include an output terminal **47**, and an input terminal **48** coupled to the drain terminal **42** of the second NMOS **40**.

[0075] Considering a scenario whereby the solid state switch **20***b* is arranged such that the voltage at the drain terminal **42** of the second NMOS **40** is greater than the voltage at the drain terminal **12** of the first NMOS **10**. The second buffer **46** which is connected to the second NMOS **40** may be arranged to provide a voltage to the bulk terminal **45** substantially equal to the voltage at the drain terminal **42** of the second NMOS **40**. This reduces the leakage current i.sub.lkg_off through parasitic diode D**2***b* as per equation (1). The second buffer **46** connected to the second NMOS **40** may also be arranged to provide a current source to the bulk terminal **45** of the second NMOS **40** based on the voltage at the drain terminal **42** of the second NMOS **40**. Thus, the addition of the second buffer **46** reduces leakage via the parasitic diode D**2***b* when compared to an arrangement of the solid state switch **20** where the bulk terminal **45** is directly electrically coupled to the source terminal **44** (e.g., without the second buffer **46** present). Therefore, the second buffer **46** may reduce (or eliminate) leakage at the drain terminal **42** of the second NMOS **40** when the voltage at the drain terminal **42** of the second NMOS **10** is greater than the voltage at the source terminal **44** of the second NMOS **40** (e.g., by providing a 0V across the parasitic diodes D**2***b*).

[0076] Advantageously, the buffers **36**, **46** of bi-directional solid state switch **20***b* may reduce (or eliminate) leakage if there is any voltage differential between the drain terminal **42** of the second NMOS **40** and the drain terminal **12** of the first NMOS **10**.

[0077] In addition, the parasitic diode D**1***a* (between a source terminal **14** and a bulk terminal **16** of the first NMOS **10**) may be reverse biased and reduces the leakage current sourced from the second buffer **46** over the forward biased parasitic diode D**2***a*. Thus, from the perspective of the input/output terminals of the solid state switch **20***b*, leakage may be eliminated, and the accuracy of component measurements connected via the solid state switch **20***b* can be improved.

[0078] Optionally, the first and/or second buffer **36**, **46** is a unitary gain buffer (UGB) (e.g., an operational amplifier-based buffer circuit with unitary gain). The first and second UGB **36**, **46** provides a low impedance output and therefore can source or sink current at the output terminal **37**, **47** of the first and second UGB **36**, **46**, respectively. Therefore, current leakage from parasitic diodes D**1***b* and D**2***b* can (in theory) be eliminated, or at least greatly reduced. Therefore, an advantage of a UGB **36**, **46** is that leakage may be reduced at the drain terminal of the first and second NMOS **10**, **40**. A reduced leakage can result in improved power efficiency and accuracy of component measurements connected via the solid state switch (e.g., solid state switch **20***a*, **20***b*, etc.). In addition, one or more UGBs **36**, **46** can beneficially provide a particularly low voltage difference between the input terminal **38**, **48** voltage and output terminal **37**, **47**. Therefore, the UGB **36**, **46** can achieve improved accuracy of component current measurements due to low current leakage at the input and/or output of the solid state switch **20***a*, **20***b*. Additionally, if one or more buffers (such as first and second buffers **36**, **46**, and optionally, gate drive circuitry configured to operate the gates of the first NMOS **10** and the second NMOS **40** each) comprise an operational amplifier, then improved THD can be achieved.

[0079] FIG. **4***b* shows a solid state switch **20***c* which may operate substantially similarly to the solid state switch **20***b* of FIG. **4***a*. The solid state switch **20***c* comprises PMOS devices **10***a*, **40***a* in place of the NMOS devices **10**, **40**. Specifically, a first terminal (e.g., a source terminal) of a first PMOS **10***a* is coupled to a first terminal (e.g., a source terminal) of a second PMOS **40***a*.

[0080] FIG. **5***a* shows a bi-directional solid state switch **49** whereby the arrangement of the first and second MOSFETs **10**, **40**, and the first and second buffers **36**, **46** corresponds with the arrangement as shown in FIG. **4***a*. Advantageously, FIG. **5***a* provides bidirectional leakage reduction. In addition, FIG. **5***a* comprises a third and a fourth MOSFETs **50**, **60**. The third MOSFET **50** is in series with the fourth MOSFET **60**. The third MOSFET **50** comprises: a gate terminal **51**, a drain terminal **52**, a source terminal **54** and a bulk terminal **65**. The fourth MOSFET **60** comprises: a gate terminal **61**, a drain terminal **62**, a source terminal **64** and a bulk terminal **65**. The output terminal **37** of the first buffer **36** is coupled to the bulk terminal **55** of the third MOSFET **50**, and the input terminal **38** of the first buffer **36** is coupled to the drain terminal **52** of the third MOSFET **50**. The output terminal **47** of the second buffer **46** is coupled to the bulk terminal **65** of the fourth MOSFET **60**, and the input terminal **48** of the second buffer **46** is coupled to the drain terminal **62** of the fourth MOSFET **60**. The third MOSFET may be a first PMOS **50** and the fourth MOSFET **60** may be a second PMOS **60**.

[0081] The solid state switch **49** of FIG. **5***a* is a bi-directional solid state transmission-gate (i.e., T-gate) switch **49**. The bi-directional solid state T-gate switch **49** comprises an n-type bi-directional solid state switch comprising first and second NMOS **10**, **40** in parallel with a p-type bi-directional solid state switch comprising first and second PMOS **50**, **60**. Advantageously, the bi-directional solid state T-gate switch **49** provides full rail-to-rail signal pass.

[0082] Alternatively, the solid state T-gate switch **49** may be arranged to only reduce (or eliminate) leakage from only one of the two input/output terminals of the solid state switch **49**, (similar to the solid state switch **20***a* of FIG. **3**) by removing the second buffer **46** (or by removing the first buffer **36**) from the solid state T-gate switch **49** shown at FIG. **5***a*.

[0083] Alternatively, a solid state switch may be substantially similar to the bi-directional solid state T-gate switch **49** of FIG. **5***a* but may only comprise one of either the first or second PMOS **50**, **60** and therefore, not provide leakage compensation to the remaining PMOS, for example, see FIG. **5***b*. Alternatively or in addition, a solid state switch may be substantially similar to the bi-directional solid state T-gate switch **49** of FIG. **5***a* but may only comprise one of either the first or second NMOS **10**, **40**. For a first example, the bi-directional solid state T-gate switch **49** of FIG. **5***a* may be modified such that any one of the MOSFETs **10**, **40**, **50**, or **60** of the bi-directional solid state T-gate switch **49** are absent. If one of the MOSFETs **10**, **40**, **50**, **60** are absent then the

resulting open-circuited source terminal (e.g., either **14**, **44**, **54**, **64**) may be coupled to: [0084] a) one or more of the remaining source terminals of the remaining MOSFETs (i.e., any three of MOSFETs **10**, **40**, **50**, **60**), as shown by FIG. **5***c*; or [0085] b) coupled to the input or output terminal **52**, **62** of the solid state switch (e.g., such that the all remaining three MOSFETs pass current between the input and the output terminals **52**, **62** of the solid state switch when the solid state switch is turned “ON”), as shown by FIG. **5***b*.
[0086] In example a), shown by FIG. **5***c*, the gate drive signal of the second MOSFET **40** is greater than the gate drive signal of the remaining MOSFETs **10**, **50**. For example, the gate drive signal of the second MOSFET **40** may be Vth (i.e., “ON” threshold voltage) above the gate drive signals of the remaining MOSFETs **10**, **50**.
[0087] For a second example, the bi-directional solid state T-gate switch **49** of FIG. **5***a* may be modified such that any two of the MOSFETs **10**, **40**, **50**, or **60** are absent. In the second example, it is preferable (in order to maintain bi-directional functionality) that either: first PMOS **50** and second PMOS **60** are absent (i.e., corresponding to solid state switch **20** of FIG. **4***a*); first NMOS **10** and second NMOS **40** are absent (i.e., corresponding to solid state switch **20***a* of FIG. **4***b*); first NMOS **10** and second PMOS **60** are absent (and the source terminal **54** of the first PMOS **50** is coupled to the source terminal **44** of the second NMOS **40**); or, first PMOS **50** and second NMOS **40** are absent (and the source terminal **14** of the first NMOS **10** is coupled to the source terminal **64** of the second PMOS **60**).
[0088] The T-gate arrangement described with reference to FIG. **5***a* comprises two buffers **36**, **46**. Alternatively, the bi-directional solid state T-gate switch **49** with may be reconfigured to use a buffer associated with the bulk terminal of each MOSFET.
[0089] Alternatively, the MOSFETs of FIGS. **5***a*-**6***b* may be any type of MOSFETS respectively.
[0090] In an example, the solid state switches as shown in FIG. **3**, **4***a*, **4***b*, **5***a*, **5***b*, or **5***c* may further comprise an electrical component for overvoltage protection. The electrical component may be arranged in series between the first and second MOSFETs (e.g., NMOSs **10** and **40** of FIGS. **3**, **4***a*, **4***b*, **5***a*, **5***b*, and **5***c*). The electrical component may protect the solid state switch in applications where overvoltage conditions can cause damage to a device to which the switch is connected. That is, the solid state switch may have overvoltage protection characteristics.
[0091] FIG. **6***a* shows a bi-directional solid state switch **68** with an overvoltage MOSFET **70** (for overvoltage protection) arranged in series between the first and second MOSFETs **10**, **40**. Although, in practice all three MOSFETS (i.e., first and second MOSFETs **10**, **40**, and overvoltage MOSFET **70**) provide overvoltage protection, specifically, NMOS devices provide overvoltage protection for positive voltages and PMOS devices provide overvoltage protection for negative voltages. Advantageously, FIG. **6***a* provides bidirectional leakage reduction, due to the presence of two buffers **36**, **46**. The overvoltage MOSFET is of a different type to the first and second MOSFETs **10**, **40**. The bi-directional solid state switch **68** includes three series-connected MOS transistors: first NMOS **10**, a PMOS **70** and second NMOS **40**. The first and third MOSFETs **10**, **40** are shown as N-type MOSFETs and the middle MOSFET **70** is shown as a P-type MOSFET in FIG. **6***a*. Alternatively, the first and third MOSFETs **10**, **40** may be P-type MOSFETs and the middle MOSFET **76** may be a N-type MOSFET. Alternatively, all three MOSFETs may be of the same type, for example, this may be advantageous in examples if the expected input/output overvoltages were different. For the architecture of FIG. **6***a*, it is the voltage rating of the MOSFETs that set the overvoltage protection level.
[0092] The bi-directional solid state switch **68** of FIG. **6***a* may provide positive and negative overvoltage protection (e.g., +55V, and −45V) when the bi-directional solid state switch **68** is powered via Vdd and Vss, and when the solid state switch is unpowered (i.e., the power supplies Vdd˜=Vss˜=gnd).
[0093] The switch input/output terminal **12** of the bi-directional solid state switch **68** is the drain terminal **12** of the first NMOS transistor **10**. The bulk terminal of the first NMOS **10** is coupled to

the output terminal **37** of the first buffer **36** via a first protection circuit **72**. The drain terminal of the PMOS **70** is coupled to the source electrode of the first NMOS **10**. The source terminal of the PMOS **70** is coupled to the source electrode of the second NMOS **40**. The switch input/output terminal **42** of the solid state switch is the drain terminal of the second NMOS transistor **40**. The bulk terminal of the second NMOS **40** is coupled to the output terminal **47** of the second buffer **46** via a second protection circuit **74**. The bulk terminal of the PMOS **70** may be configured to float when the solid state switch is turned OFF′. Specifically, a bulk-source PMOS **76** with a gate terminal coupled to the gate terminal of the PMOS **70** may be arranged to cause the bulk terminal of the PMOS **70** to float when it is in an off-state. When the bi-directional solid state switch **68** is in an off-state, the bulk terminals of the first NMOS **10** and second NMOS **40** are coupled to output terminals **37**, **47** of the first and second buffers **36**, **46** respectively. Thus, reducing (or eliminating) current leakage at the inputs/outputs **12**, **42** of the bi-directional solid state switch **68**. Alternatively, the bulk-source PMOS **76** of FIG. **6***a* may be replaced with a bulk-drain PMOS **76** (i.e., between the bulk terminal and the drain terminal of the PMOS **70**).

[0094] The bulk terminal of the PMOS **70** may be configured to be coupled to the source terminal of the PMOS **70** when the bi-directional solid state switch **68** is turned 'ON', such that the PMOS **70** does not impede the signal through the bi-directional solid state switch **68**. Specifically, the bulk-source PMOS **76** with a gate terminal coupled to the gate terminal of the PMOS **70** may be arranged to cause the bulk terminal of the PMOS **70** to be coupled to the source terminal of the PMOS **70** when it is in an on-state. When the bi-directional solid state switch **68** is in an on-state, the bulk terminals of the first NMOS **10** and second NMOS **40** may be configured to float, for example, due to the first and second protection circuits **72**, **74**.

[0095] The gate terminals of the NMOSs **10**, **40** are coupled together and to a first output of a control circuit **78**. The gate terminal of the PMOS **70** is coupled to a second output of the control circuit **78**, such that the PMOS **70**, first NMOS **10**, and second NMOS **40** may be in the on-state (or off-state) at the same time (e.g., switched simultaneously).

[0096] In the example described, control signals output from the first output and second output of the control circuit **78** respectively are either +15 volts or −15 volts (alternatively, these may +5V or −5V), derived from the supply voltages. The control signals are complementary, such that when one is +15 volts, the other is-15 volts. When the first control signal is +15 volts, the first and second NMOSs **10**, **40** are turned on. The second control signal will at that time be −15 volts, and this voltage on the gate terminal **32** of the PMOS **70** (and bulk-source PMOS **76**) will turn PMOS **70** on. Thus, all three in-series MOSFETs (i.e., first NMOS **10**, PMOS **70**, and second NMOS **40**) are turned 'ON', and the switch is "closed", meaning that an input signal can be transmitted from the input to the output of the bi-directional solid state switch **68**.

[0097] When the second control signal is +15 volts and the first control signal is −15 volts, all three MOSFETs (i.e., first NMOS **10**, PMOS **70**, and second NMOS **40**) will be turned 'OFF', and bi-directional solid state switch **68** will open, that is, its resistance from the input terminal **12** to the output terminal **42** will be extremely high (typically megohms), effectively preventing signal transmission through the bi-directional solid state switch **68**.

[0098] FIG. **6***b*, shows an alternative bi-directional solid state switch **68***a* with an overvoltage MOSFET **70** (for overvoltage protection) arranged in series between the first and second MOSFETs **10**, **40**. The bi-directional solid state switch **68***a* shows all of the components of the bi-directional solid state switch **68** of FIG. **6***a*. The same reference numerals are used to denote the same/corresponding features in relation to FIG. **6***a* and will not be described in detail again below. Specifically, in contrast to the bi-directional solid state switch **68** of FIG. **6***a*, the first and second protection circuits **72**, **74** may be configured to be always 'ON' by tying the gates of NMOS devices to Vdd and the gates of PMOS devices to Vss, as shown in FIG. **6***b*. This will always drive the bulk terminals of all three MOSFETS (i.e., first NMOS **10**, PMOS **70**, and second NMOS **40**) with the buffer **36**, **46** except in an overvoltage event when the bulk terminals will be automatically floated. Advantageously, this provides better performance in the 'ON' state of the bi-directional solid state switch **68***a*.

[0099] Each MOSFET of the solid state switch **20***a*, **20***b*, **20***c*, **49**, **68** may be a low voltage MOSFET. Low voltage may be defined as: less than or equal to 5V, or, less than or equal to 10V. A low voltage MOS may be configured operate with a Vds which is less than or equal to 5V, or, less than or equal to 10V. Low voltage components and MOSFET devices may advantageously provide reduced area and power consumption in comparison to higher voltage components.

[0100] The values of any components herein may be changed depending on the application and/or designer choice.

[0101] For all of the above designs and circuits, it is possible to add additional components while still achieving the technical effects associated with each embodiment.

[0102] Alternatively, the MOS devices herein may comprise an isolation terminal or may not comprise an isolation terminal (e.g. Silicon on Insulator (SOI) devices). The output terminal of the first buffer **36** may only be coupled to the bulk terminal of the first MOSFET **10** when the first MOSFET **10** is in the off-state. The output terminal of the second buffer **46** may only be coupled to the bulk terminal of the second MOSFET **40** when the second MOSFET **40** is in the off-state.

[0103] The first and second MOSFET devices **10**, **40** of FIGS. **3**-**6***b* are shown as n-type devices, however, the teachings can be readily applied to p-type devices by a person skilled in the art, and vice versa.

[0104] The output terminals **37** of the first and second buffers **36**, **46** are shown to be coupled to the bulk terminals of both the first MOSFET **10** and the second MOSFET **40** in FIGS. **3**-**6***b*. Alternatively, any means for providing a current source to the bulk terminal of the first MOSFET **10** based on the voltage at the drain terminal **12** of the first MOSFET **10** may be used in order to achieve reduced leakage, and/or any means for providing a current source to the bulk terminal of the second MOSFET **40** based on the voltage at the drain terminal **42** of the second MOSFET **40** may be used in order to achieve reduced leakage. For example, if the second MOSFET **40** has different leakage characteristics to the first MOSFET **10**. The above applies equally to the other MOSFETs in a similar arrangement, e.g., the third and Fourth MOSFETs **50**, **60** of FIGS. **5***a*-**5***c*, and/or PMOS **70** of FIGS. **6***a*, **6***b*.

[0105] The MOSFETs herein (e.g., **10**, **40**, **50**, **60**, **70**) may be any type of MOSFET design.

[0106] When reference is made to a drain, source, gate, bulk, buried layer, isolation layer or other input/output of a component, this may include a drain terminal, source terminal, gate terminal, bulk terminal, buried layer terminal, isolation terminal or other input/output terminal of a component, respectively (and vice versa).

[0107] Throughout the described embodiments, where a MOSFET has a 'drain terminal', this may be referred to as a 'first terminal', and where a MOSFET has a 'source terminal', this may be referred to as a 'second terminal'. This is due to the symmetrical nature of certain MOSFETs as a result of their fabrication method. Additionally, it will be understood that due to the symmetrical nature of MOSFETs, in any embodiment described herein, the orientation of the first and second terminals of each MOSFET may be arbitrary. For example, FIG. **4***a* may show the first terminal of the first MOSFET **10** coupled in series to the second terminal of the second MOSFET **40**, and the input terminals of the first and second buffers **36**, **46** may be coupled to the second terminal of the first MOSFET **10** and the first terminal of the second MOSFET **40** respectively.

[0108] The first and/or second buffer **36**, **46** may each be a unity gain buffer in FIGS. **3** to **6***b*. The unity gain buffer may comprise an operational amplifier to reduce leakage and capacitance at the drain of the connected MOS (e.g., the first MOSFET **10** in FIGS. **3** to **6***b*). The operational amplifier may be one of: a continuous time auto-zero operational amplifier; or a continuous time ping-pong auto zero operational amplifier, to achieve a low voltage difference across the input terminal **38**, **48** and the output terminal **37**, **47** of the first and/or second buffer **36**, **46**.

[0109] Alternatively, the first and/or second buffer **36**, **46** may each be a voltage follower circuit.

The voltage follower circuit can reduce capacitance at the drain of the connected MOSFET (e.g., the first MOSFET **10** in FIGS. **3** to **6***b*). The voltage follower circuit is simpler to implement and reduces circuit complexity. The voltage follower circuit is usually an open loop or feedforward type circuit.
[0110] Alternatively, the first and/or second buffer **36**, **46** may each be a cascade complementary source follower (CCSF). The CCSF can reduce capacitance at the drain of the connected MOSFET (e.g., the first MOSFET **10** in FIGS. **3** to **6***b*). The CCSF is simpler to implement and reduces circuit complexity. The CCSF can achieve near zero voltage drop across the CCSF, although, this may vary with temperature and silicon processes. Alternatively, the first and/or second buffer **36**, **46** may each be any other implementation of a source follower type circuit or buffer.
[0111] The first and/or second buffer **36**, **46** may each be any means for providing a current source to the bulk terminal of a MOSFET (e.g., MOSFET **10**, and/or MOSFET **40** respectively) based on the voltage at the first terminal (e.g., the drain terminal) of the MOSFET.
[0112] In an example, the NMOS devices **10**, **40**, **50**, **60** and/or the PMOS devices **10***a*, **20***a*, **70** may be a non-isolated MOS devices. Alternatively, the NMOS devices **10**, **10***a*, **40**, **40***a*, **50**, **60**, **70** may be an isolated MOS devices.
[0113] In addition to MOS devices described above, other Field Effect Transistor (FET) devices may be used in place of the MOS devices described with reference to FIGS. **1** to **6***b*. A FET device may be configured to function as a switch and is susceptible to parasitic capacitance and current leakage which can negatively impact measurement performance of equipment connected via said FET device. To overcome the impact of parasitic capacitance and current leakage, additional circuitry can be coupled to the FET device configured to function as a switch. FET devices (e.g., JFET, Silicon Carbide FET (SIC), and MOS devices, etc.) which comprise at least one parasitic diodes D**1***a* and D**1***b* (as shown with reference to FIGS. **1***a* and **1***b*) can benefit from the additional circuitry described with reference to FIGS. **1** to **6***b*. Examples of other FET devices with additional circuitry corresponding to certain examples shown in FIGS. **1** to **6***b* are described below.
[0114] FIGS. **7***a* and **7***b* show bi-directional solid state switches **20***d*, **20***e* with unidirectional leakage reduction. Some of the components in FIGS. **7***a* and **7***b* are similar to components in FIG. **3** and use the same reference numbers. For purposes of conciseness, similar components will not be described in detail again. In contrast to the solid state switch device **20***a* of FIG. **3**, solid state switch device **20***d*, **20***e* of FIGS. **7***a* and **7***b* show a first and second FET **10***b*, **10***c*, **40***b*, **40***c* as an alternative to the first and second NMOS **10**, **40**.
[0115] In the example of FIG. **7***a*, the first and second FETs are first and second SiCs **10***b*, **40***b* (e.g., in a MOSFET form as show, or JFET form (not shown)). In the example of FIG. **7***b*, the first and second FETs are first and second JFETs **10***c*, **40***c*. The bulk terminal (also called a back gate terminal) of a JFET may also be the gate terminal of a JFET. In another example, the first and second FETs may be any other type of FET.
[0116] In an example, the first and second FETs may be first and second SiCs as will be described below. FIG. **7***a* shows the bi-directional solid state switch **20***d* with unidirectional leakage reduction comprising the first SiC **10***b* coupled to a first buffer **36**. The first SiC **10***b* being configured to be switched between an on-state (i.e., switched ‘ON’) and an off-state (i.e., switched ‘OFF’). The first buffer **36** includes an input terminal **38** coupled to the drain terminal **12***a* of the first SiC **10***b*. The first buffer **36** is arranged to provide a voltage to the bulk terminal **16***a* substantially equal to the voltage at the drain terminal **12***a* of the first SiC **10***b*. This reduces the leakage current i.sub.lkg_off through parasitic diode D**3***b*. The first buffer **36** connected to the first SiC **10***b* may also be arranged to provide a current source to the bulk terminal **16***a* of the first SiC **10***b* based on the voltage at the drain terminal **12***a* of the first SiC **10***b*.
[0117] A mirrored arrangement of the first SiC **10***b* is provided by a second SiC **40***b*. The first SiC **10***b* may be substantially similar to the second SiC **40***b*. Thus, when the voltage at the drain terminal **12***a* of the first SiC **10***b* is greater than the voltage at the drain terminal **42***a* of the second

SiC **40***b*, current leakage through the solid state switch **20***d* when it is in an off-state may be reduced (or eliminated). The solid state switch **20***d* is constructed by coupling the source terminals **14***a*, **44***a* of each first SiC **10***b* and second SiC **40***b* together (either directly or optionally, via an electrical component for overvoltage protection). This arrangement ensures that the parasitic diode D**4***a* (between a source terminal **44***a* and a bulk terminal **45***a* of the second SiC **40***b*) may be reverse biased and reduces the leakage current sourced from the first buffer **36** over the forward biased parasitic diode D**3***a*. There may be a small current leakage via the reverse biased parasitic diode D**4***a* (i.e., sourced from the first buffer **36**), however, advantageously, this is not 'seen' at (i.e., does not interact with) the drain terminal of the first SiC **10***b*. Thus, from the perspective of the input terminal **12***a* of the solid state switch **20***d*, current leakage may be eliminated.

[0118] Specifically, FIG. **7***a* shows the solid state switch **20***d* comprises the second SiC **40***b*. The second SiC **40***b* being configured to be switched between an on-state and an off-state. FIG. **7***a* shows the second SiC **40***b* with parasitic diodes D**4***a* and D**4***b*. The second SiC **40***b* comprises a gate terminal **41***a*, a drain terminal **42***a*, a source terminal **44***a*, and a bulk terminal **45***a*. The bulk terminal **45***a* is accessible to a circuit designer.

[0119] An input of the solid state switch **20***d* may be the drain terminal **12***a* of the first SiC **10***b*. An output of the solid state switch **20***d* may be the drain terminal **42***a* of the second SiC **40***b*.

[0120] A switching signal coupled to the gate **41***a* of the second SiC **40***b* may also be coupled to the gate **11***a* of the first SiC **10***b*, such that the first and second SiCs **10***b*, **40***b* are configured to be in the same switching state, either 'ON' or 'OFF'. Thus, during an off-state of the solid state switch **20***d*, the first and second SiCs **10***b* and **40***b* can be switched 'OFF' simultaneously. Similarly, during an on-state of the solid state switch **20***d*, the first and second SiCs **10***b* and **40***b* can be switched 'ON' simultaneously.

[0121] In an off-state, the bulk terminal **45***a* of second SiC **40***b* may be coupled to the most negative supply of the switch (e.g., 0V, gnd, or Vss). In an on-state, the bulk terminal **45***a* of second SiC **40***b* may also be coupled to the most negative supply of the switch (e.g., 0V, gnd, or Vss). Alternatively, in an on-state, the bulk terminal **45***a* of second SiC **40***b* may be back gate switched. In back gate switching, the bulk terminal (e.g., **45***b*) is tied to the signal passing through the switch (e.g., second SiC **40***b*) to reduce the body effect and improve performance.

[0122] FIG. **8** shows a bi-directional solid state switch **49***c* with bi-directional leakage reduction. Some of the components in FIG. **8** are similar to components in FIG. **5***a* and use the same reference numbers. For purposes of conciseness, similar components will not be described in detail again. In contrast to the solid state switch device **49** of FIG. 5*a*, solid state switch device **49***c* of FIG. **8** shows a first, second, third, and fourth FET **10***b*, **40***b*, **50***a*, **60***a* as an alternative to the first, second, third, and fourth MOSFETs **10**, **40**, **50**, **60**.

[0123] In an example, the first, second, third, and fourth FETs **10***b*, **40***b*, **50***a*, **60***a* may be first, second, third, and fourth SiCs, JFETs, other type of FETs, or combinations thereof.

[0124] In an example, the first, second, third, and fourth FETs **10***b*, **40***b*, **50***a*, **60***a* may be first, second, third, and fourth SiCs as will be described below. FIG. **8** shows a bi-directional solid state switch **49***c*. Advantageously, the example of FIG. **8** provides bidirectional leakage reduction. In addition, FIG. **8** comprises a third SiC **50***a* and a fourth SiC **60***a*. The third SiC **50***a* is in series with the fourth SiC **60***a*. The third SiC **50***a* comprises: a gate terminal **51***a*, a drain terminal **52***a*, a source terminal **54***a* and a bulk terminal **65***a*. The fourth SiC **60***a* comprises: a gate terminal **61***a*, a drain terminal **62***a*, a source terminal **64***a* and a bulk terminal **65***a*. The output terminal **37** of the first buffer **36** is coupled to the bulk terminal **55***a* of the third SiC **50***a*, and the input terminal **38** of the first buffer **36** is coupled to the drain terminal **52***a* of the third SiC **50***a*. The output terminal **47** of the second buffer **46** is coupled to the bulk terminal **65***a* of the fourth SiC **60***a*, and the input terminal **48** of the second buffer **46** is coupled to the drain terminal **62***a* of the fourth SiC **60***a*. The third SiC **50***a* may be p-type and the fourth SiC **60***a* may be p-type.

[0125] The solid state switch **49***c* of FIG. **8** is a bi-directional solid state transmission-gate (i.e., T-gate) switch **49***c*. The bi-directional solid state T-gate switch **49***c* comprises an n-type bi-directional solid state switch comprising first and second n-type SiCs **10***b*, **40***b* in parallel with a p-type bi-directional solid state switch comprising first and second n-type SiCs **50***a*, **60***a*. Advantageously, the bi-directional solid state T-gate switch **49***c* provides full rail-to-rail signal pass.

[0126] FIG. **9** shows a bi-directional solid state switch **68***b* with an overvoltage FET **70***a* (for overvoltage protection) arranged in series between the first and second FETs **10***b*, **40***b*. Some of the components in FIG. **9** are similar to components in FIG. **6***a* and use the same reference numbers. For purposes of conciseness, similar components will not be described in detail again. In contrast to the solid state switch device **68** of FIG. **6***a*, solid state switch device **68***b* of FIG. **9** shows a first, second, and overvoltage FET **10***b*, **40***b*, **70***a* as an alternative to the first, second, and overvoltage MOSFETs **10**, **40**, **70**. The overvoltage FET **70***a* is of a different type to the first and second FETs **10***b*, **40***b*.

[0127] In an example, the first, second, and overvoltage FETs **10***b*, **40***b*, **70***a* may be first, second, and overvoltage JFETs, SiCs, or other type of FET.

[0128] In an example, the first, second, and overvoltage FETs **10***b*, **40***b*, **70***a* may be first, second, and overvoltage SiCs as will be described below. FIG. **9** provides bidirectional leakage reduction, due to the presence of two buffers **36**, **46**. The overvoltage SiC **70***a* is of a different type to the first and second SiCs **10***b*, **40***b*. The bi-directional solid state switch **68***b* includes three series-connected SiC transistors: first n-type SiC **10***b*, an overvoltage p-type SiC **70***a* and second n-type SiC **40***b*. The first and third SiCs **10***b*, **40***b* are shown as n-type SiCs and the middle SiC **70***a* is shown as a p-type SiC in FIG. **9**. Alternatively, the first and third SiCs **10***b*, **40***b* may be p-type SiCs and the middle SiC **70***a* may be a n-type SiC. Alternatively, all three SiCs may be of the same type, for example, this may be advantageous in examples if the expected input/output overvoltages were different. For the architecture of FIG. **9**, it is the voltage rating of the FETs that set the overvoltage protection level.

[0129] The bi-directional solid state switch **68***b* of FIG. **9** may provide positive and negative overvoltage protection (e.g., +55V, and −45V) when the bi-directional solid state switch **68***b* is powered via Vdd and Vss, and when the solid state switch is unpowered (i.e., the power supplies Vdd˜=Vss˜=gnd). The switch input/output terminal **12***a* of the bi-directional solid state switch **68***b* is the drain terminal **12***a* of the first SiC transistor **10***b*. The bulk terminal of the first SiC **10***b* is coupled to the output terminal **37** of the first buffer **36** via a first protection circuit **72**. The drain terminal of the overvoltage SiC **70***s* is coupled to the source electrode of the first SiC **10***b*. The source terminal of the overvoltage SiC **70***a* is coupled to the source electrode of the second SiC **40***b*. The switch input/output terminal **42***a* of the solid state switch is the drain terminal of the second SiC transistor **40***b*. The bulk terminal of the second SiC **40***b* is coupled to the output terminal **47** of the second buffer **46** via a second protection circuit **74**. The bulk terminal of the overvoltage SiC **70***a* may be configured to float when the solid state switch is turned ‘OFF’. Specifically, a bulk-source SiC **76***a* with a gate terminal coupled to the gate terminal of the overvoltage SiC **70***a* may be arranged to cause the bulk terminal of the overvoltage SiC **70***a* to float when it is in an off-state. When the bi-directional solid state switch **68***b* is in an off-state, the bulk terminals of the first SiC **10***b* and second SiC **40***b* are coupled to output terminals **37**, **47** of the first and second buffers **36**, **46** respectively. Alternatively, the bulk-source SiC **76***a* of FIG. **9** may be replaced with a bulk-drain SiC (i.e., between the bulk terminal and the drain terminal of the overvoltage SiC **70***a*). In an example, the bulk-source SiC (or the bulk-drain SiC) may be any FET.

[0130] The gate terminals of the SiCs **10***b*, **40***b* are coupled together and to a first output of a control circuit **78**. The gate terminal of the overvoltage SiC **70***a* is coupled to a second output of the control circuit **78**, such that the overvoltage SiC **70***a*, first SiC **10***b*, and second SiC **40***b* may be in the on-state (or off-state) at the same time (e.g., switched simultaneously).

General

[0131] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” “include,” “including,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.”

[0132] The words “coupled” or “connected”, as generally used herein, refer to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the Detailed Description using the singular or plural number may also include the plural or singular number, respectively. The words “or” in reference to a list of two or more items, is intended to cover all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

[0133] It is to be understood that one or more features from one or more of the above-described embodiments may be combined with one or more features of one or more other ones of the above-described embodiments, so as to form further embodiments which are within the scope of the appended claims.

Numbered Clauses

[0134] Numbered clause 1. A solid state switch, comprising: [0135] a first metal-oxide-semiconductor field-effect transistor, MOSFET, comprising: a first terminal, a second terminal, a bulk terminal and a gate terminal, and configured to be switched between an on-state and an off-state; [0136] a second MOSFET in series with the first MOSFET, wherein the second MOSFET comprises a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first MOSFET is connected to the second terminal of the second MOSFET; and, [0137] a first buffer comprising an output terminal coupled to the bulk terminal of the first MOSFET, and an input terminal coupled to the first terminal of the first MOSFET.

[0138] Numbered clause 2. The solid state switch of numbered clause 1, wherein the first buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.

[0139] Numbered clause 3. The solid state switch of numbered clause 1 or 2, further comprising a second buffer comprising an output terminal coupled to the bulk terminal of the second MOSFET, and an input terminal coupled to the first terminal of the second MOSFET.

[0140] Numbered clause 4. The solid state switch of numbered clause 3, wherein the second buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.

[0141] Numbered clause 5. The solid state switch of any previous numbered clause, wherein the first buffer is a UGB so as to reduce leakage at the first terminal of the first MOSFET.

[0142] Numbered clause 6. The solid state switch of any previous numbered clause, wherein the first MOSFET is a low voltage MOSFET, and the second MOSFET is a low voltage MOSFET.

[0143] Numbered clause 7. The solid state switch of any previous numbered clause, wherein the first MOSFET is an isolated MOSFET, and the second MOSFET is an isolated MOSFET.

[0144] Numbered clause 8. The solid states switch of any previous numbered clause, wherein the first terminal of the first MOSFET is a drain terminal, wherein the first terminal of the second MOSFET is a drain terminal, wherein the second terminal of the first MOSFET is a source terminal, wherein the second terminal of the second MOSFET is a source terminal.

[0145] Numbered clause 9. The solid state switch of any preceding numbered clause, wherein the first and second MOSFET are both a first-type MOSFET, wherein the first-type is n-type or p-type.

[0146] Numbered clause 10. The solid state switch of numbered clause 9, further comprising an electrical component for overvoltage protection, wherein the electrical component is arranged in series between the first and second MOSFETs.

[0147] Numbered clause 11. The solid state switch of numbered clause 10, wherein the electrical component is a third MOSFET, wherein the third MOSFET is a second-type MOSFET, wherein the second-type is p-type or n-type, wherein the first-type is different to the second-type.

[0148] Numbered clause 12. The solid state switch of numbered clauses 10 or 11, wherein the electrical component protects the solid state switch in applications where overvoltage conditions can cause damage to a device to which the switch is connected.
[0149] Numbered clause 13. The solid state switch of any of numbered clauses 1 to 9, wherein the solid state switch is a transmission-gate switch comprising the first MOSFET in parallel with a third MOSFET, wherein the third MOSFET, comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the third MOSFET is configured to be switched between an on-state and an off-state, wherein the third MOSFET is a first-type MOSFET, wherein the first MOSFET is a second-type MOSFET, wherein the first-type is n-type or p-type, wherein the second-type is p-type or n-type, wherein the first-type is different to the second-type.
[0150] Numbered clause 14. The solid state switch of numbered clause 13, wherein the solid state switch further comprises a fourth MOSFET in parallel with the second MOSFET, wherein the fourth MOSFET comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the fourth MOSFET is configured to be switched between an on-state and an off-state, wherein the fourth MOSFET is a third-type MOSFET, wherein the second MOSFET is a fourth-type MOSFET, wherein the third-type is n-type or p-type, wherein the fourth-type is p-type or n-type, wherein the third-type is different to the fourth-type.
[0151] Numbered clause 15. The solid state switch of numbered clause 14, [0152] wherein the fourth MOSFET is in series with the third MOSFET, and wherein the second terminal of the third MOSFET is connected to the second terminal of the fourth MOSFET.
[0153] Numbered clause 16. The solid state switch of numbered clause 15, wherein the output terminal of the first buffer is coupled to the bulk terminal of the third MOSFET, and the input terminal of the first buffer is coupled to the first terminal of the third MOSFET; and/or when dependent on numbered clause 3, [0154] wherein the output terminal of the second buffer is coupled to the bulk terminal of the fourth MOSFET, and the input terminal of the second buffer is coupled to the first terminal of the fourth MOSFET.
[0155] Numbered clause 17. The solid states switch of numbered clause 16, wherein the first terminal of the third MOSFET is a drain terminal, wherein the first terminal of the fourth MOSFET is a drain terminal, wherein the second terminal of the third MOSFET is a source terminal, wherein the second terminal of the fourth MOSFET is a source terminal.
[0156] Numbered clause 18. The solid state switch of any of numbered clauses 13 to 17, wherein the third MOSFET is a low voltage MOSFET, and the fourth MOSFET is a low voltage MOSFET.
[0157] Numbered clause 19. The solid state switch of any of numbered clauses 13 to 18, wherein the third MOSFET is an isolated MOSFET, and the fourth MOSFET is an isolated MOSFET.
[0158] Numbered clause 20. A solid state switch, comprising: [0159] a first metal-oxide-semiconductor field-effect transistor, MOSFET, comprising: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and configured to be switched between an on-state and an off-state; [0160] a second MOSFET in series with the first MOSFET, wherein the second comprises: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first MOSFET is coupled to the second terminal of the second MOSFET; [0161] means for providing a current source to the bulk terminal of the first MOSFET based on the voltage at the first terminal of the first MOSFET; and means for providing a current source to the bulk terminal of the second MOSFET based on the voltage at the first terminal of the second MOSFET.
[0162] Numbered clause 21. A solid state switch, comprising: [0163] a first field-effect transistor, FET, comprising: a first terminal, a second terminal, a bulk terminal and a gate terminal, and configured to be switched between an on-state and an off-state; [0164] a second FET in series with the first FET, wherein the second FET comprises a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first FET is connected to the second terminal of the second FET; and, [0165] a first buffer comprising an output terminal coupled to the bulk terminal of the first FET, and an input terminal coupled to the first terminal of

the first FET.
[0166] Numbered clause 22. The solid state switch of numbered clause 21, wherein the first buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.
[0167] Numbered clause 23. The solid state switch of numbered clause 21 or 22, further comprising a second buffer comprising an output terminal coupled to the bulk terminal of the second FET, and an input terminal coupled to the first terminal of the second FET.
[0168] Numbered clause 24. The solid state switch of numbered clause 23, wherein the second buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.
[0169] Numbered clause 25. The solid state switch of any of numbered clauses 21 to 24, wherein the first buffer is a UGB so as to reduce leakage at the first terminal of the first FET.
[0170] Numbered clause 26. The solid state switch of any of numbered clauses 21 to 25, wherein the first FET is a low voltage FET, and the second FET is a low voltage FET.
[0171] Numbered clause 27. The solid state switch of any of numbered clauses 21 to 26, wherein the first FET is an isolated FET, and the second FET is an isolated FET.
[0172] Numbered clause 28. The solid states switch of any of numbered clauses 21 to 27, wherein the first terminal of the first FET is a drain terminal, wherein the first terminal of the second FET is a drain terminal, wherein the second terminal of the first FET is a source terminal, wherein the second terminal of the second FET is a source terminal.
[0173] Numbered clause 29. The solid state switch of any of numbered clauses 21 to 28, wherein the first and second FET are both a first-type FET, wherein the first-type is n-type or p-type.
[0174] Numbered clause 30. The solid state switch of numbered clause 29, further comprising an electrical component for overvoltage protection, wherein the electrical component is arranged in series between the first and second FETs.
[0175] Numbered clause 31. The solid state switch of numbered clause 30, wherein the electrical component is a third FET, wherein the third FET is a second-type FET, wherein the second-type is p-type or n-type, wherein the first-type is different to the second-type.
[0176] Numbered clause 32. The solid state switch of any of numbered clauses 30 or 31, wherein the electrical component protects the solid state switch in applications where overvoltage conditions can cause damage to a device to which the switch is connected.
[0177] Numbered clause 33. The solid state switch of any of numbered clauses 21 to 29, wherein the solid state switch is a transmission-gate switch comprising the first FET in parallel with a third FET, wherein the third FET, comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the third FET is configured to be switched between an on-state and an off-state, wherein the third FET is a first-type FET, wherein the first FET is a second-type FET, wherein the first-type is n-type or p-type, wherein the second-type is p-type or n-type, wherein the first-type is different to the second-type.
[0178] Numbered clause 34. The solid state switch of numbered clause 33, wherein the solid state switch further comprises a fourth FET in parallel with the second FET, wherein the fourth FET comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the fourth FET is configured to be switched between an on-state and an off-state, wherein the fourth FET is a third-type FET, wherein the second FET is a fourth-type FET, wherein the third-type is n-type or p-type, wherein the fourth-type is p-type or n-type, wherein the third-type is different to the fourth-type.
[0179] Numbered clause 35. The solid state switch of numbered clause 34, wherein the fourth FET is in series with the third FET, and wherein the second terminal of the third FET is connected to the second terminal of the fourth FET.
[0180] Numbered clause 36. The solid state switch of numbered clause 35, wherein the output terminal of the first buffer is coupled to the bulk terminal of the third FET, and the input terminal of the first buffer is coupled to the first terminal of the third FET, and/or when dependent on

numbered clause 23, wherein the output terminal of the second buffer is coupled to the bulk terminal of the fourth FET, and the input terminal of the second buffer is coupled to the first terminal of the fourth FET.
[0181] Numbered clause 37. The solid states switch of numbered clause 36, wherein the first terminal of the third FET is a drain terminal, wherein the first terminal of the fourth FET is a drain terminal, wherein the second terminal of the third FET is a source terminal, wherein the second terminal of the fourth FET is a source terminal.
[0182] Numbered clause 38. The solid state switch of any of numbered clauses 33 to 37, wherein the third FET is a low voltage FET, and the fourth FET is a low voltage FET.
[0183] Numbered clause 39. The solid state switch of any of numbered clauses 33 to 38, wherein the third FET is an isolated FET, and the fourth FET is an isolated FET.
[0184] Numbered clause 40. The solid state switch of any of numbered clauses 21 to 39, wherein the first FET is a JFET or a Silicon Carbide FET (SiC); and/or wherein the second FET is a JFET or a SiC.
[0185] Numbered clause 41. A solid state switch, comprising: [0186] a first field-effect transistor, FET, comprising: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and configured to be switched between an on-state and an off-state; [0187] a second FET in series with the first FET, wherein the second comprises: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first FET is coupled to the second terminal of the second FET; [0188] means for providing a current source to the bulk terminal of the first FET based on the voltage at the first terminal of the first FET; and [0189] means for providing a current source to the bulk terminal of the second FET based on the voltage at the first terminal of the second FET.

Claims

1. A solid state switch, comprising: a first field-effect transistor, FET, comprising: a first terminal, a second terminal, a bulk terminal and a gate terminal, and configured to be switched between an on-state and an off-state; a second FET in series with the first FET, wherein the second FET comprises a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first FET is connected to the second terminal of the second FET; and, a first buffer comprising an output terminal coupled to the bulk terminal of the first FET, and an input terminal coupled to the first terminal of the first FET.
2. The solid state switch of claim 1, wherein the first buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.
3. The solid state switch of claim 1, further comprising a second buffer comprising an output terminal coupled to the bulk terminal of the second FET, and an input terminal coupled to the first terminal of the second FET.
4. The solid state switch of claim 3, wherein the second buffer is: a unity gain buffer, UGB; a voltage follower; or a cascade complementary source follower.
5. The solid state switch of claim 1, wherein the first buffer is a UGB so as to reduce leakage at the first terminal of the first FET.
6. The solid state switch of claim 1, wherein the first FET is a low voltage FET, and the second FET is a low voltage FET.
7. The solid state switch of claim 1, wherein the first FET is an isolated FET, and the second FET is an isolated FET.
8. The solid states switch of claim 1, wherein the first terminal of the first FET is a drain terminal, wherein the first terminal of the second FET is a drain terminal, wherein the second terminal of the first FET is a source terminal, wherein the second terminal of the second FET is a source terminal.
9. The solid state switch of claim 1, wherein the first and second FET are both a first-type FET,

wherein the first-type is n-type or p-type.
10. The solid state switch of claim 9, further comprising an electrical component for overvoltage protection, wherein the electrical component is arranged in series between the first and second FETs.
11. The solid state switch of claim 10, wherein the electrical component is a third FET, wherein the third FET is a second-type FET, wherein the second-type is p-type or n-type, wherein the first-type is different to the second-type.
12. The solid state switch of claim 10, wherein the electrical component protects the solid state switch in applications where overvoltage conditions can cause damage to a device to which the switch is connected.
13. The solid state switch of claim 1, wherein the solid state switch is a transmission-gate switch comprising the first FET in parallel with a third FET, wherein the third FET, comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the third FET is configured to be switched between an on-state and an off-state, wherein the third FET is a first-type FET, wherein the first FET is a second-type FET, wherein the first-type is n-type or p-type, wherein the second-type is p-type or n-type, wherein the first-type is different to the second-type.
14. The solid state switch of claim 13, wherein the solid state switch further comprises a fourth FET in parallel with the second FET, wherein the fourth FET comprises: a first terminal; a second terminal; a bulk terminal; and, a gate terminal, and the fourth FET is configured to be switched between an on-state and an off-state, wherein the fourth FET is a third-type FET, wherein the second FET is a fourth-type FET, wherein the third-type is n-type or p-type, wherein the fourth-type is p-type or n-type, wherein the third-type is different to the fourth-type.
15. The solid state switch of claim 14, wherein the fourth FET is in series with the third FET, and wherein the second terminal of the third FET is connected to the second terminal of the fourth FET.
16. The solid state switch of claim 15, wherein the output terminal of the first buffer is coupled to the bulk terminal of the third FET, and the input terminal of the first buffer is coupled to the first terminal of the third FET.
17. The solid states switch of claim 16, wherein the first terminal of the third FET is a drain terminal, wherein the first terminal of the fourth FET is a drain terminal, wherein the second terminal of the third FET is a source terminal, wherein the second terminal of the fourth FET is a source terminal.
18. The solid state switch of claim 1, wherein the first FET is a JFET or a Silicon Carbide FET, SiC.
19. The solid state switch of claim 1, wherein the second FET is a JFET or a Silicon Carbide FET, SiC.
20. A solid state switch, comprising: a first field-effect transistor, FET, comprising: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and configured to be switched between an on-state and an off-state; a second FET in series with the first FET, wherein the second comprises: a first terminal, a second terminal, a bulk terminal, and a gate terminal, and wherein the second terminal of the first FET is coupled to the second terminal of the second FET; means for providing a current source to the bulk terminal of the first FET based on the voltage at the first terminal of the first FET; and means for providing a current source to the bulk terminal of the second FET based on the voltage at the first terminal of the second FET.